

Agentic AI Project Synopsis

1. Project Title: VentureLens: A Multi-Agent Agentic AI System for Startup Idea Validation

2. Team Members: Ajay Teli (230937), Shruti Negi ([230844]), Diksha Balodi ([230872])

3. Problem Definition

- Validating a startup idea normally needs market research, competitor analysis, financial feasibility checks, and risk assessment — usually done manually or by a single AI model working sequentially, which is slow and prone to one-sided bias.
- Student entrepreneurs, early-stage founders, and college incubation cells who want a quick, evidence-backed first opinion on a business idea before committing time or capital.

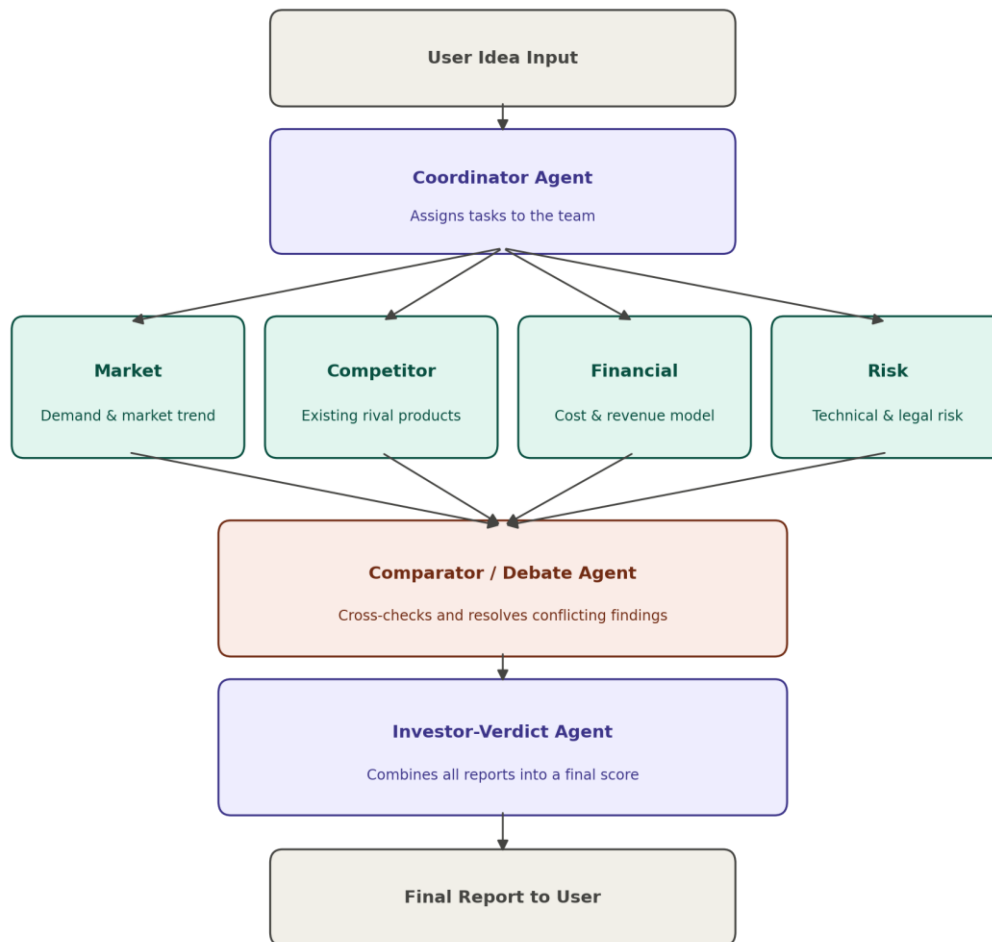
4. Objectives & Expected Outcomes

- Build a 7-agent CrewAI pipeline that automates idea validation through parallel research; reduce single-agent bias with a dedicated debate/cross-verification step; deliver a clear, scored Go/No-Go recommendation through a simple web interface.
- A working prototype, VentureLens, that takes a plain-text business idea and returns a research-backed score (out of 10) with a Go/No-Go recommendation and justification within a few minutes.

5. Proposed Methodology

- Coordinator (assigns and manages tasks); Market Research, Competitor Analysis, Financial Feasibility, and Risk Assessment agents (parallel research); Comparator/Debate agent (resolves conflicting findings); Investor-Verdict agent (final score and recommendation).
- Serper.dev web search API, used by the Market, Competitor, and Risk agents for live research.
- CrewAI (hierarchical multi-agent process), with a multi-provider LLM fallback chain for reliability, including Gemini flash-lite models and the OpenAI/llama-3.1-8b-instant model accessed via an OpenRouter API key.

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6. Feasibility & Resources

- Live web search results (Google-indexed, via Serper.dev) for market, competitor, and risk research - no proprietary dataset is required.
- Serper.dev search API; Gemini API and OpenRouter API for LLM inference.
- Python 3.10+, CrewAI, LiteLLM, Streamlit.
- A standard laptop with no GPU - all reasoning runs through cloud LLM APIs, so only lightweight local Python dependencies are needed.

7. Team Members & Individual Understanding

Member	Role/Responsibility	Individual Understanding
Ajay Teli (230937)	Team Lead -System Architecture, Agent Design & Development	Understands the full pipeline: how the coordinator delegates work, how each agent's role and tools fit together, and how the debate and fallback logic keep verdicts reliable.
Shruti Negi ([230844])	Research & Prompt Design - Market, Competitor and Risk agents	Understands how each research agent's prompt and search tool shape its findings, and how those findings are cross-verified by the Comparator agent.
Diksha Balodi ([230872])	Testing, Documentation & UI	Understands how to validate outputs against expected Go/No-Go outcomes and how the Streamlit interface presents the final score and recommendation.

8. Project Timeline

Activity	Week(s)	Deliverable
Problem & Design	Week 1	Finalised problem statement, 7-agent architecture, and system design document
Development /Coding	Weeks 2–3	Working agents.py, tasks.py and crew.py implementing all 7 agents and the hierarchical pipeline
Integration & Testing/Debugging	Week 4	End-to-end pipeline tested on sample ideas with provider fallback verified
Evaluation & Refinement	Week 5	Outputs verified against benchmark ideas (strong, weak and borderline) and prompts refined
Presentation/Interface UI	Weeks 5–6	Polished Streamlit interface (VentureLens) showing score, recommendation and justification

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Activity	Week(s)	Deliverable
Deployment	Week 6	Locally deployed, demo-ready application
Final Demo	Week 6	Live demonstration of the complete validation pipeline to evaluators

9. Team Responsibility Matrix

Activity	Ajay Teli	Shruti Negi	Diksha Balodi
Agent Design	Lead		Support
Agent/Tool Development	Lead	Support	
Integration & Testing	Support	Lead	
Evaluation & Documentation		Lead	Support
Presentation/Interface UI	Support		Lead
Deployment		Support	Lead